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(57) **ABSTRACT**

Disclosed is an air cleaner. The air cleaner includes a case including a suction port disposed on a peripheral surface of the case, an air cleaning module including a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port, and a booster module disposed above the air cleaning module. The booster module including a head cover, a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction.

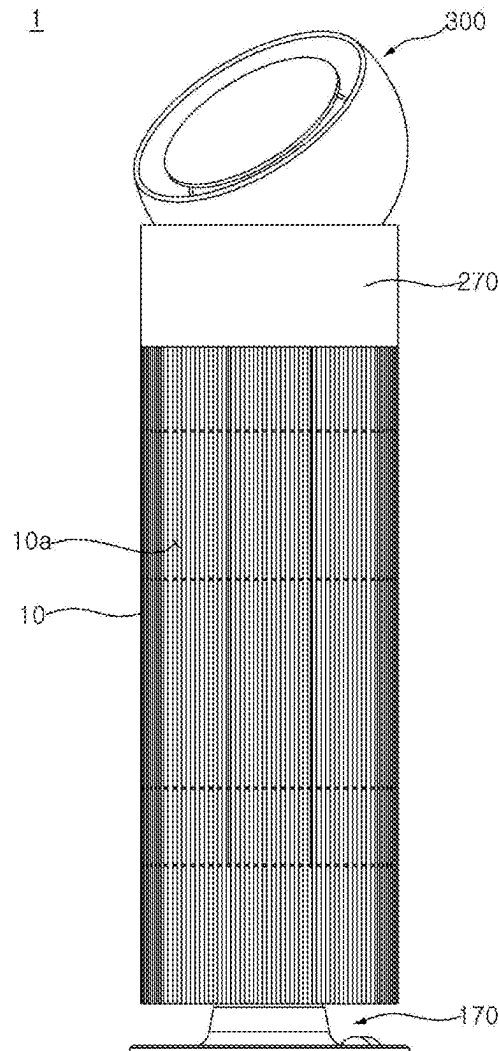


FIG. 1

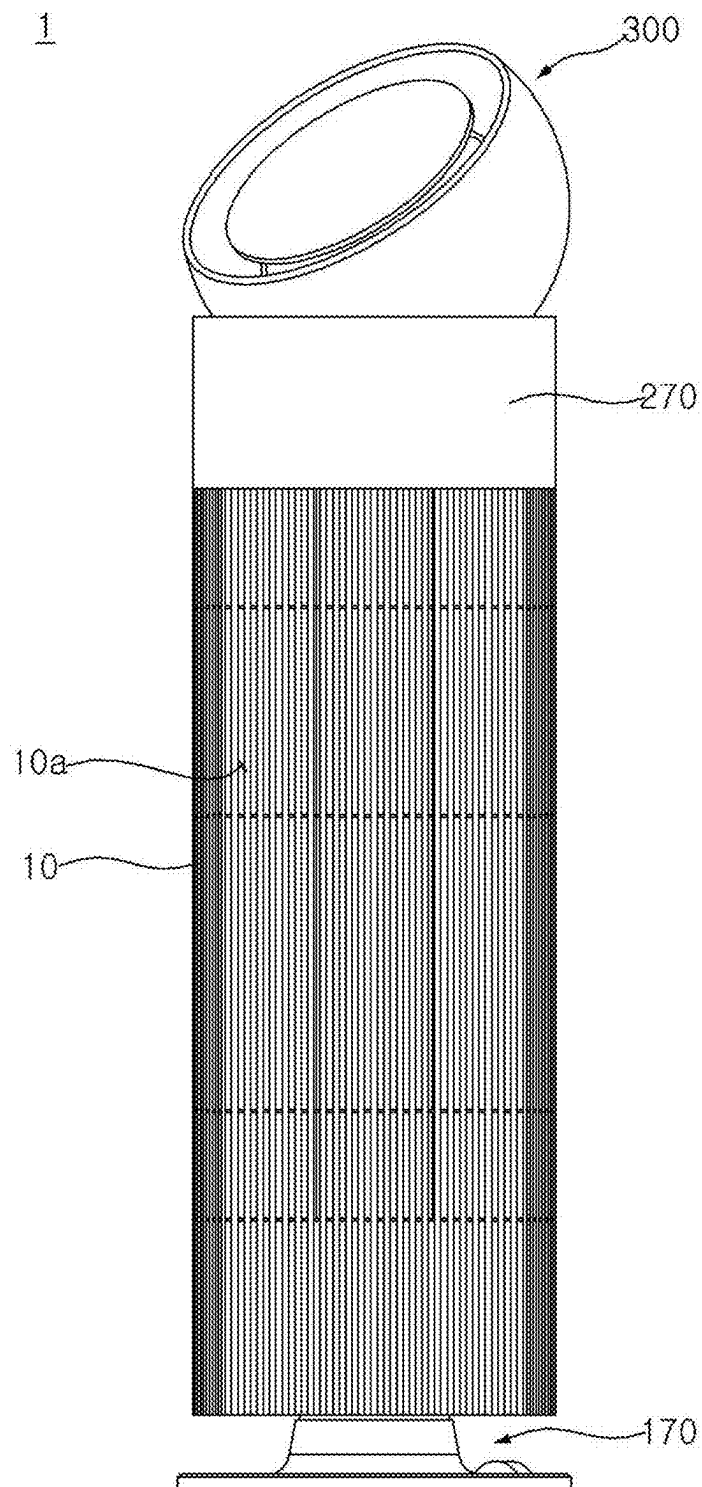


FIG. 2a

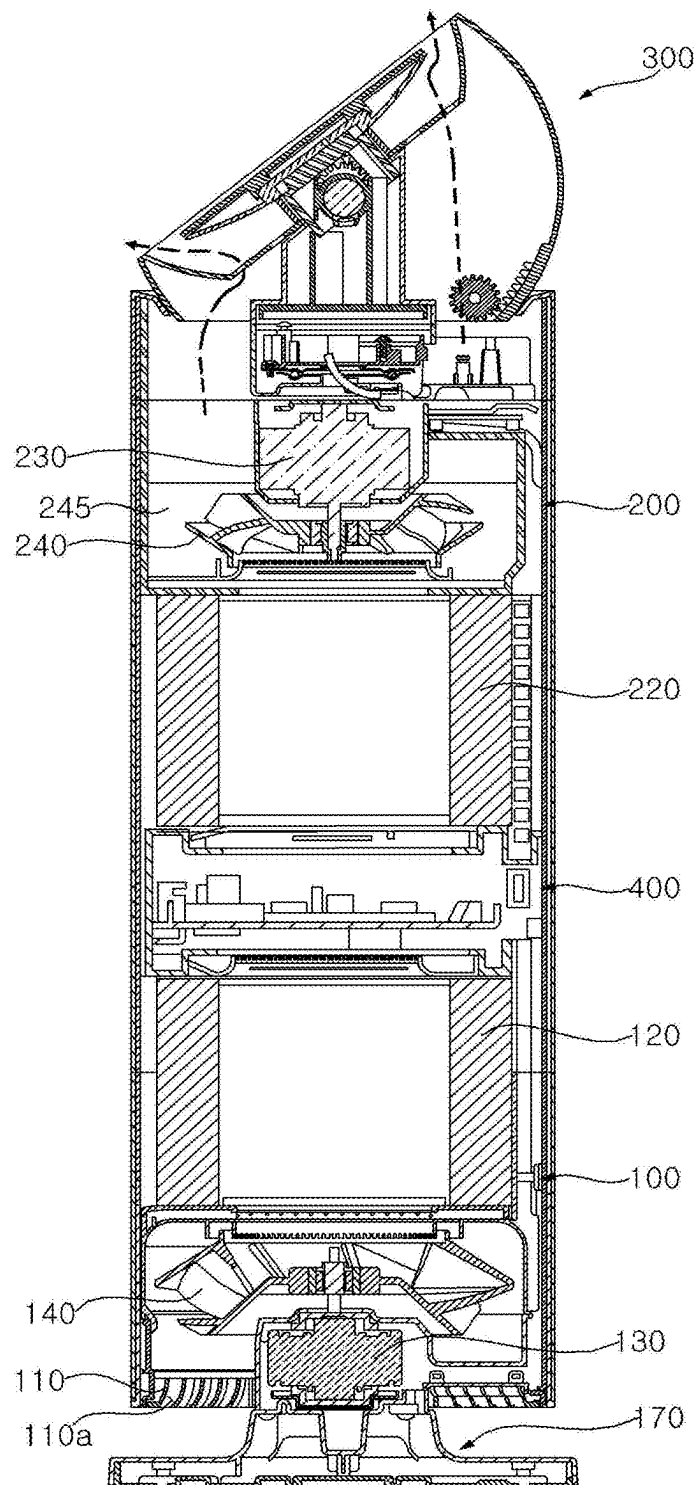


FIG. 3

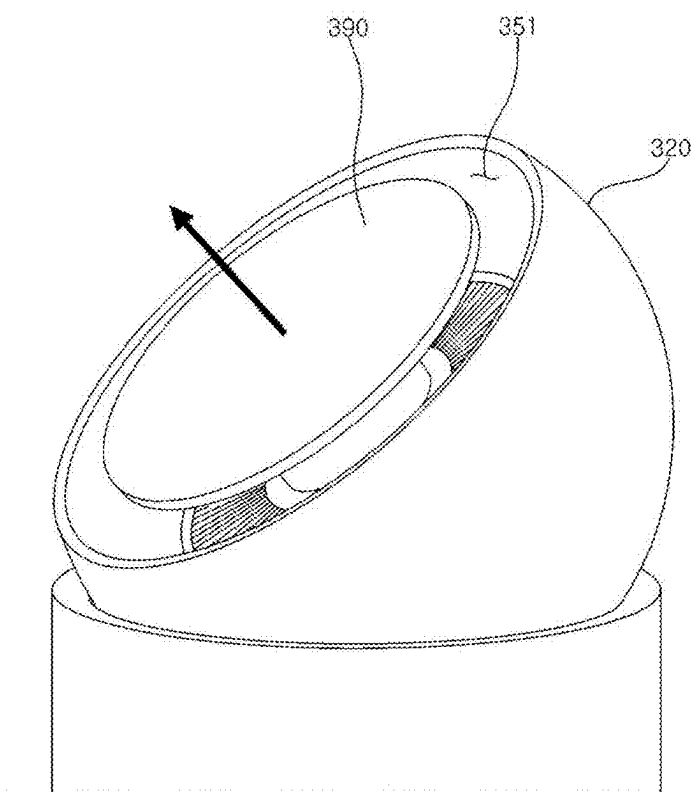


FIG. 4

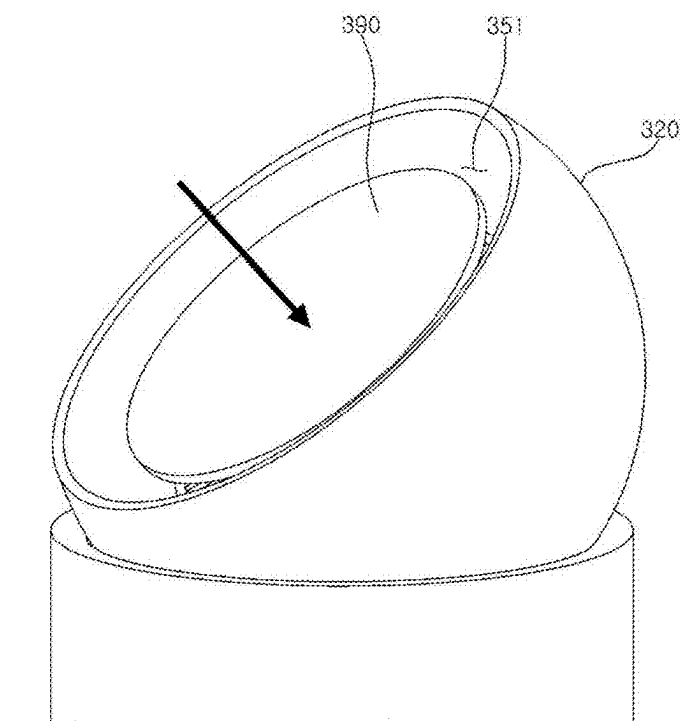


FIG. 5

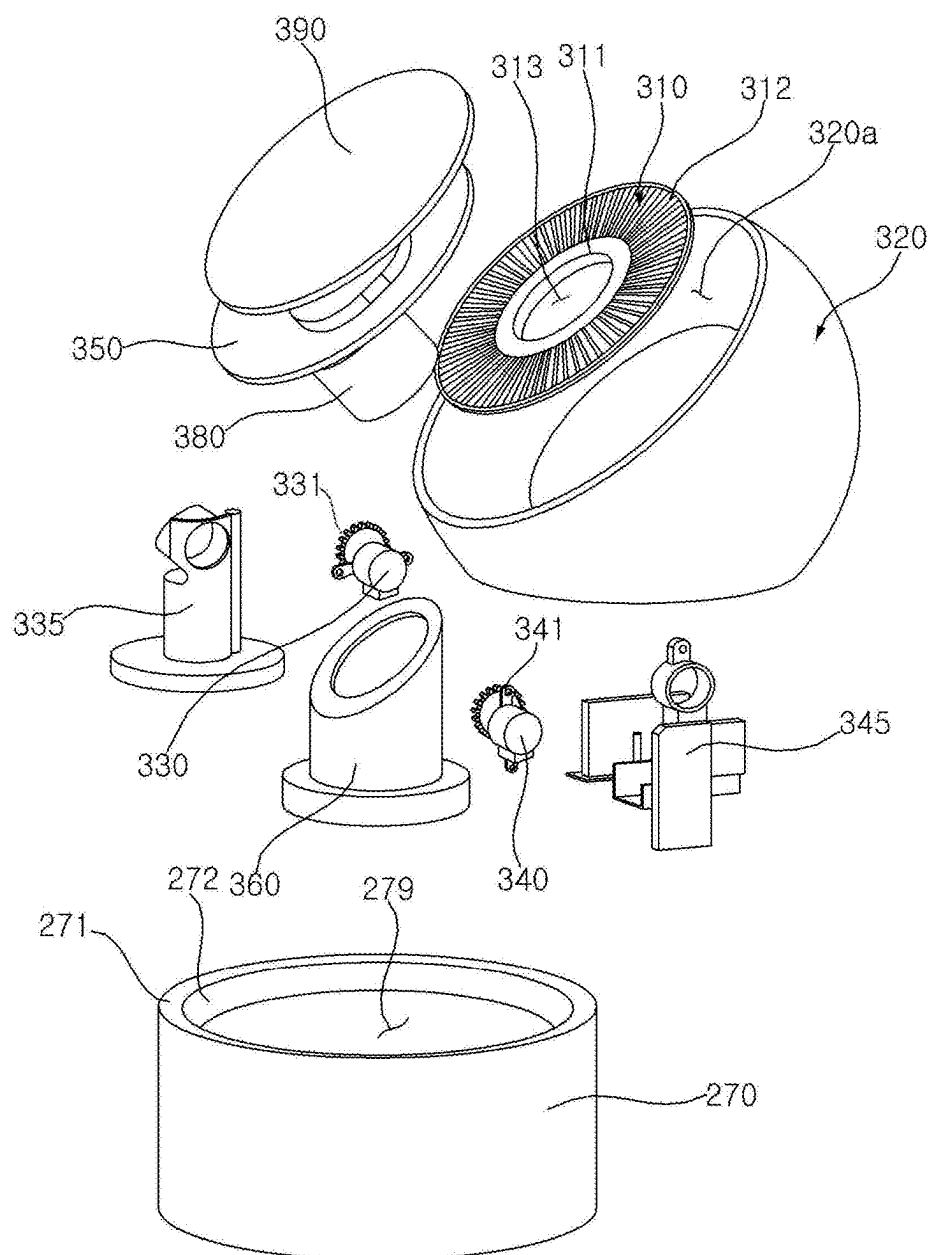


FIG. 6

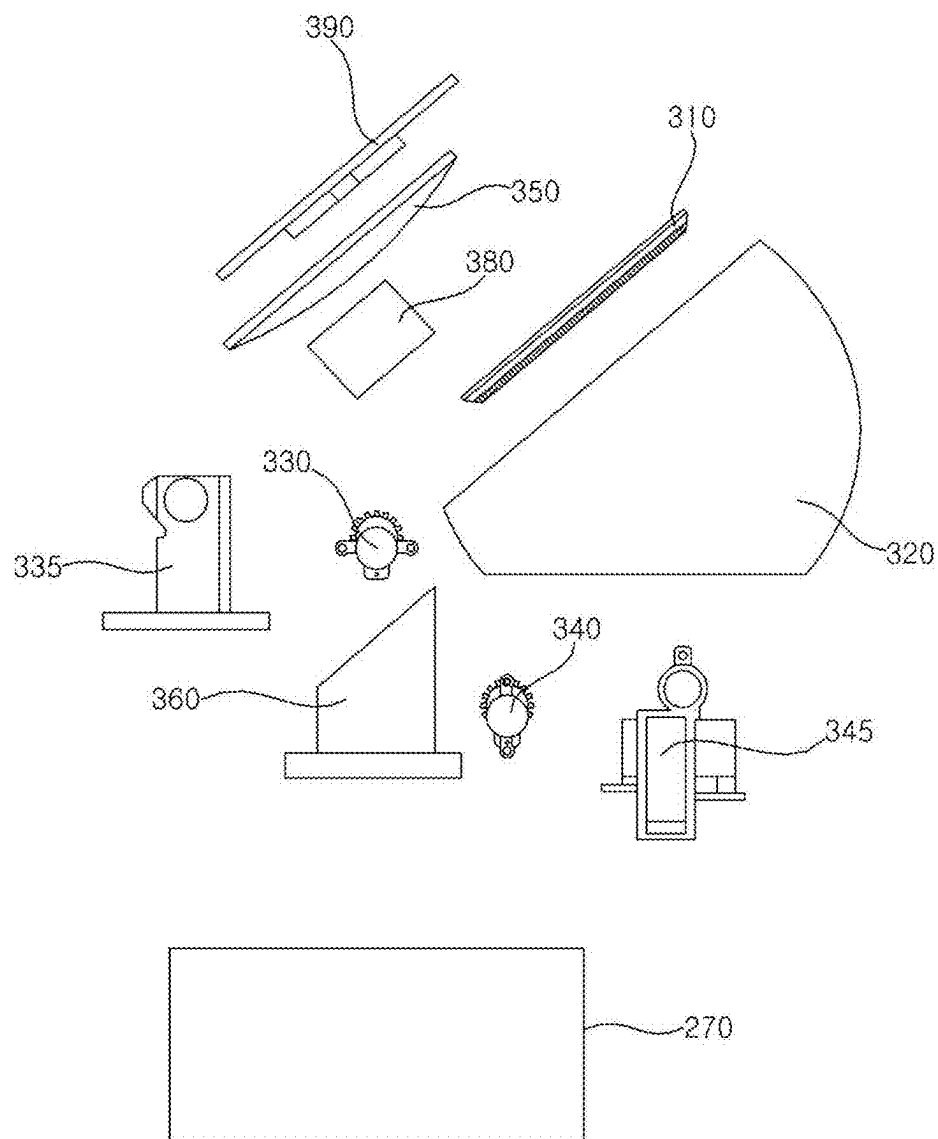


FIG. 8

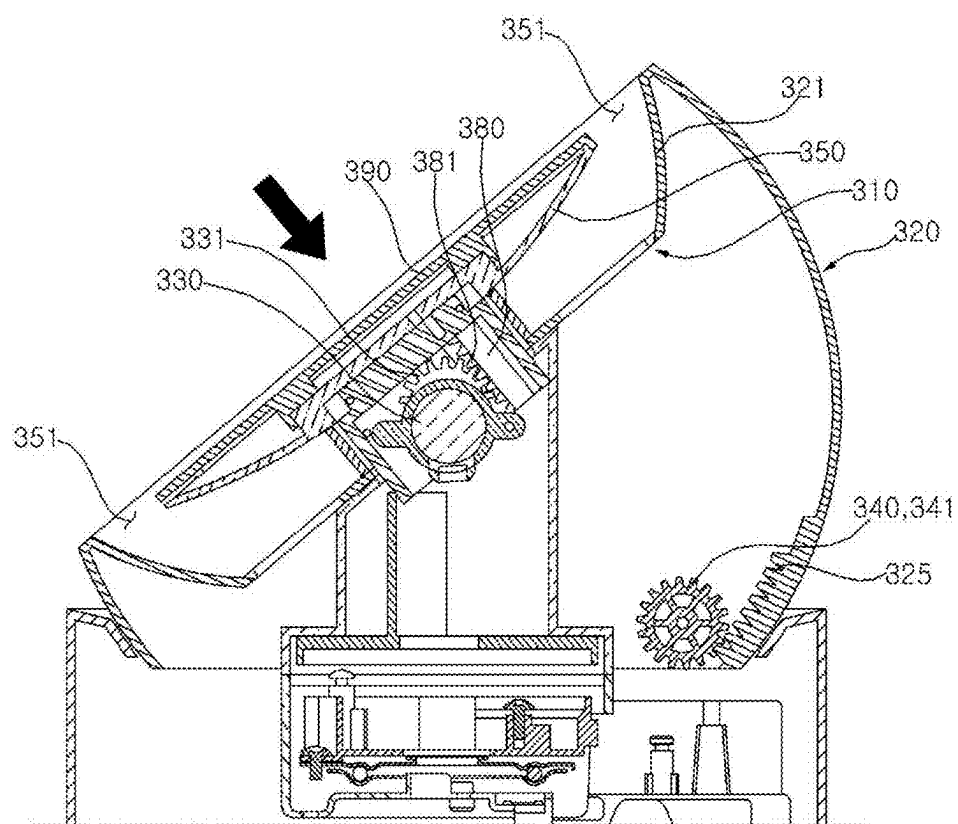


FIG. 9

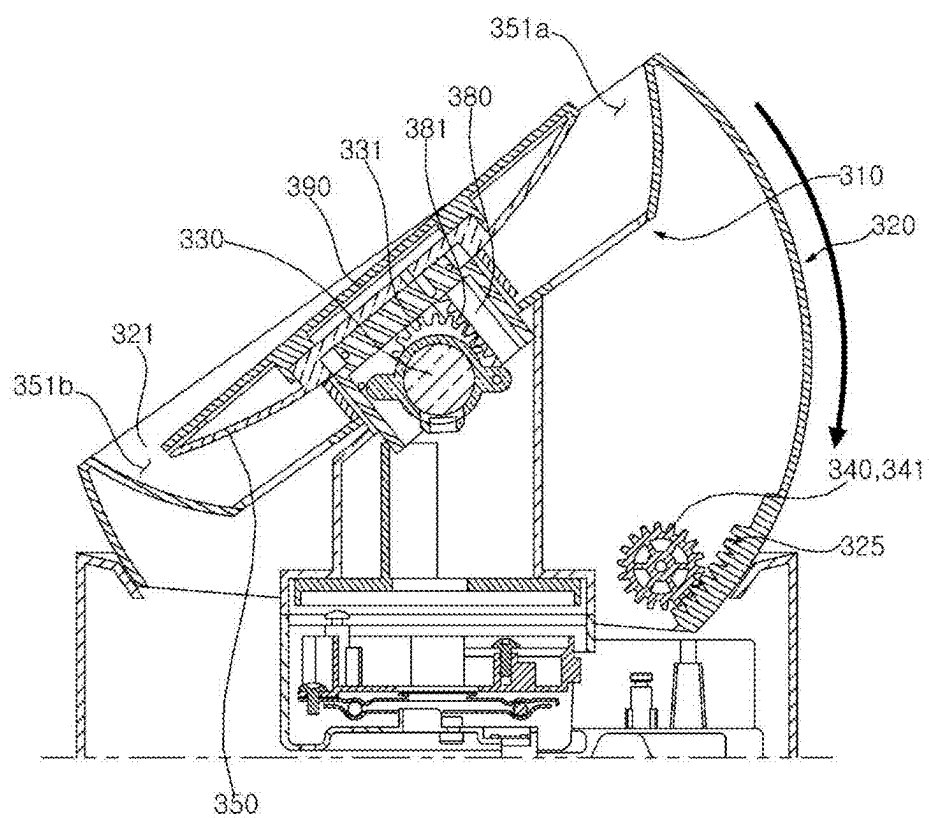


FIG. 11

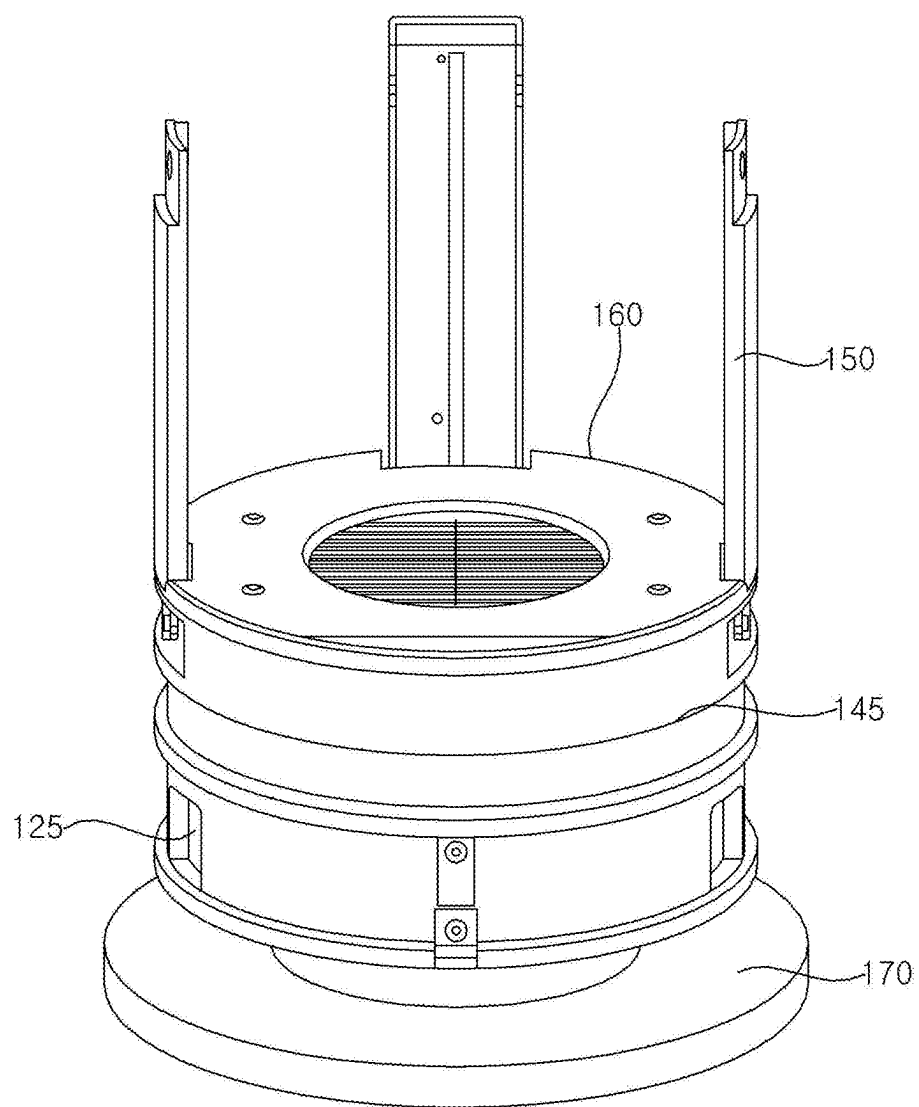


FIG. 12

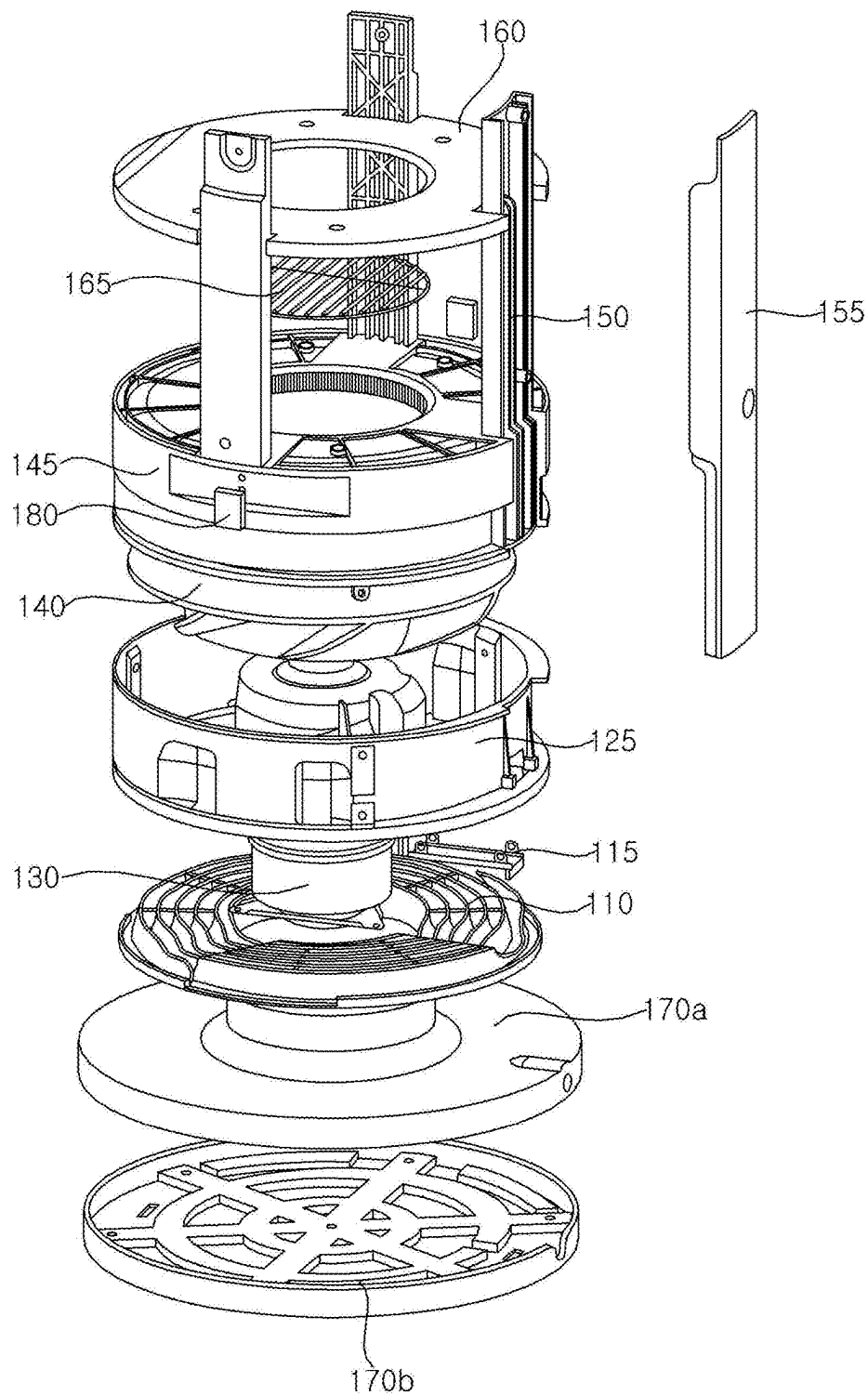


FIG. 13

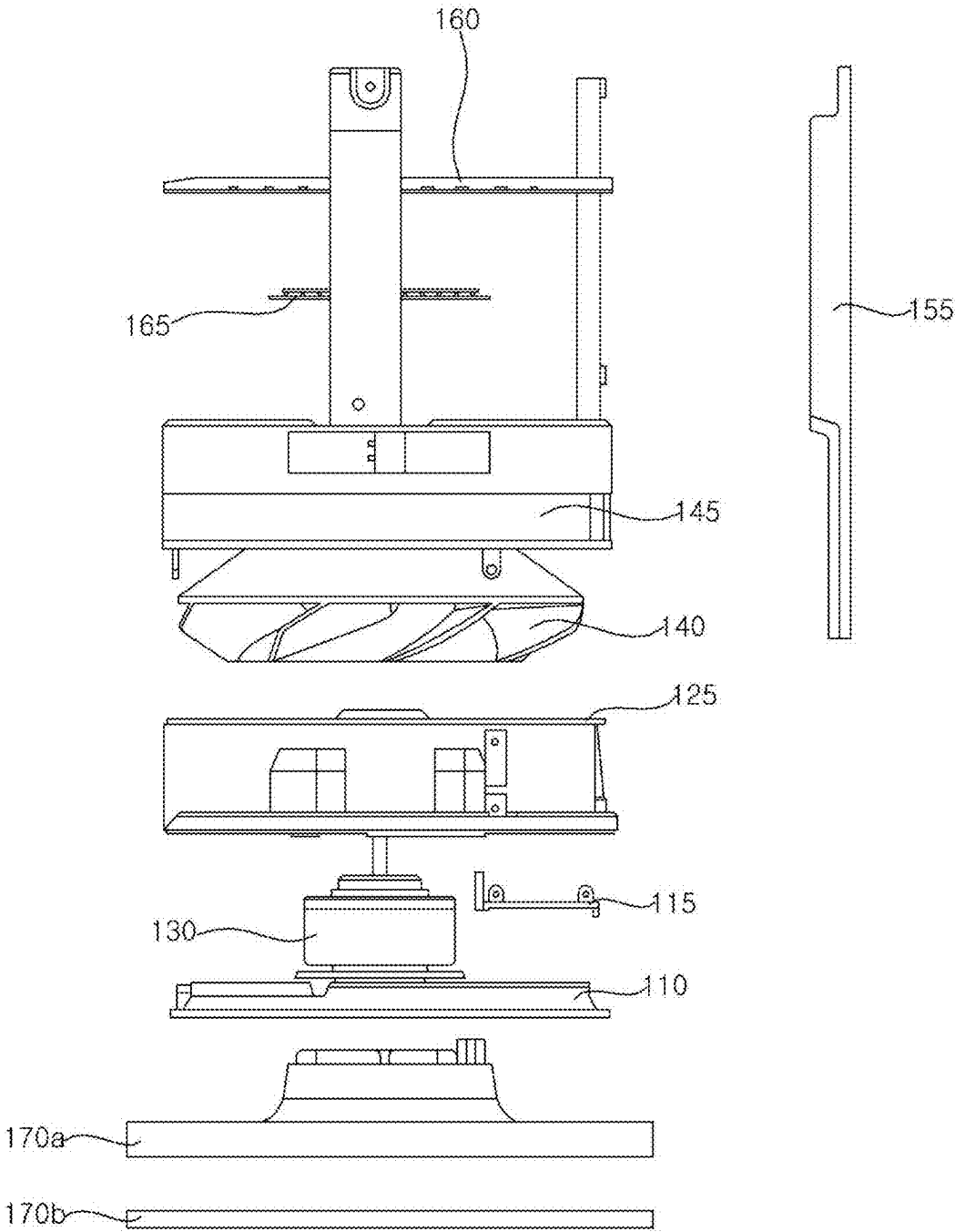


FIG. 14

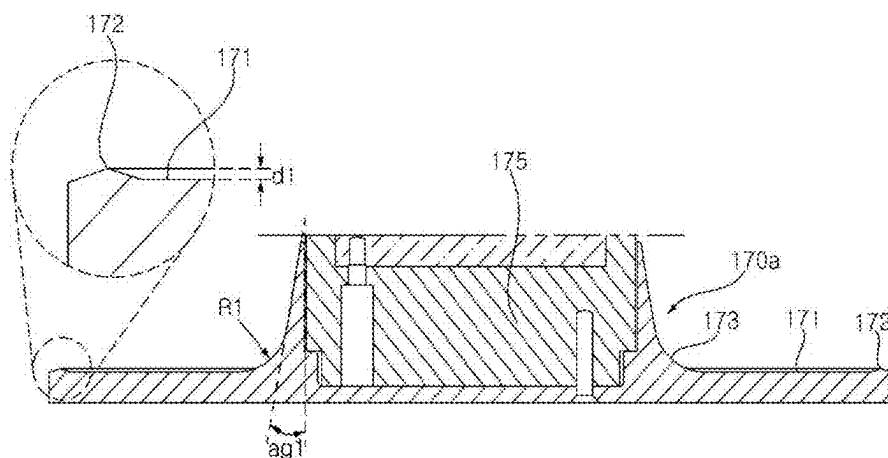


FIG. 15

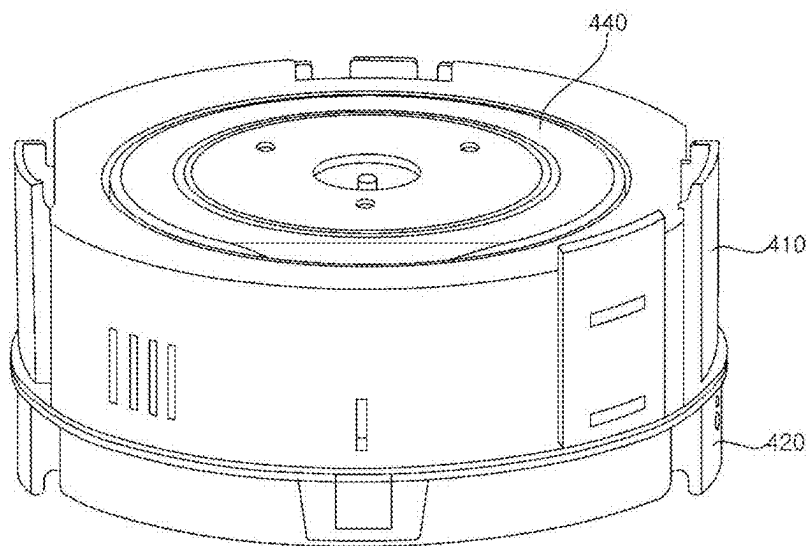


FIG. 16

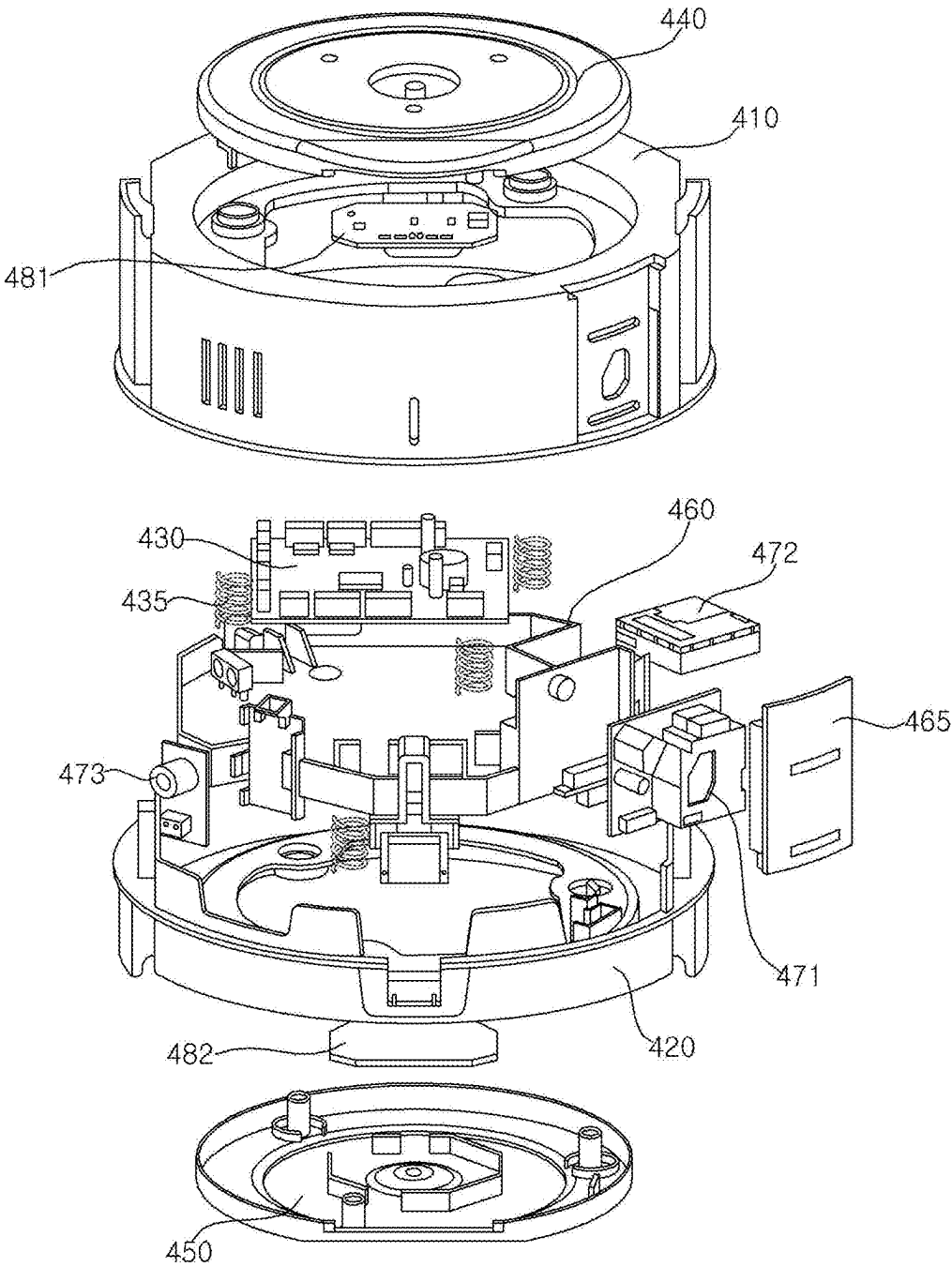


FIG. 17

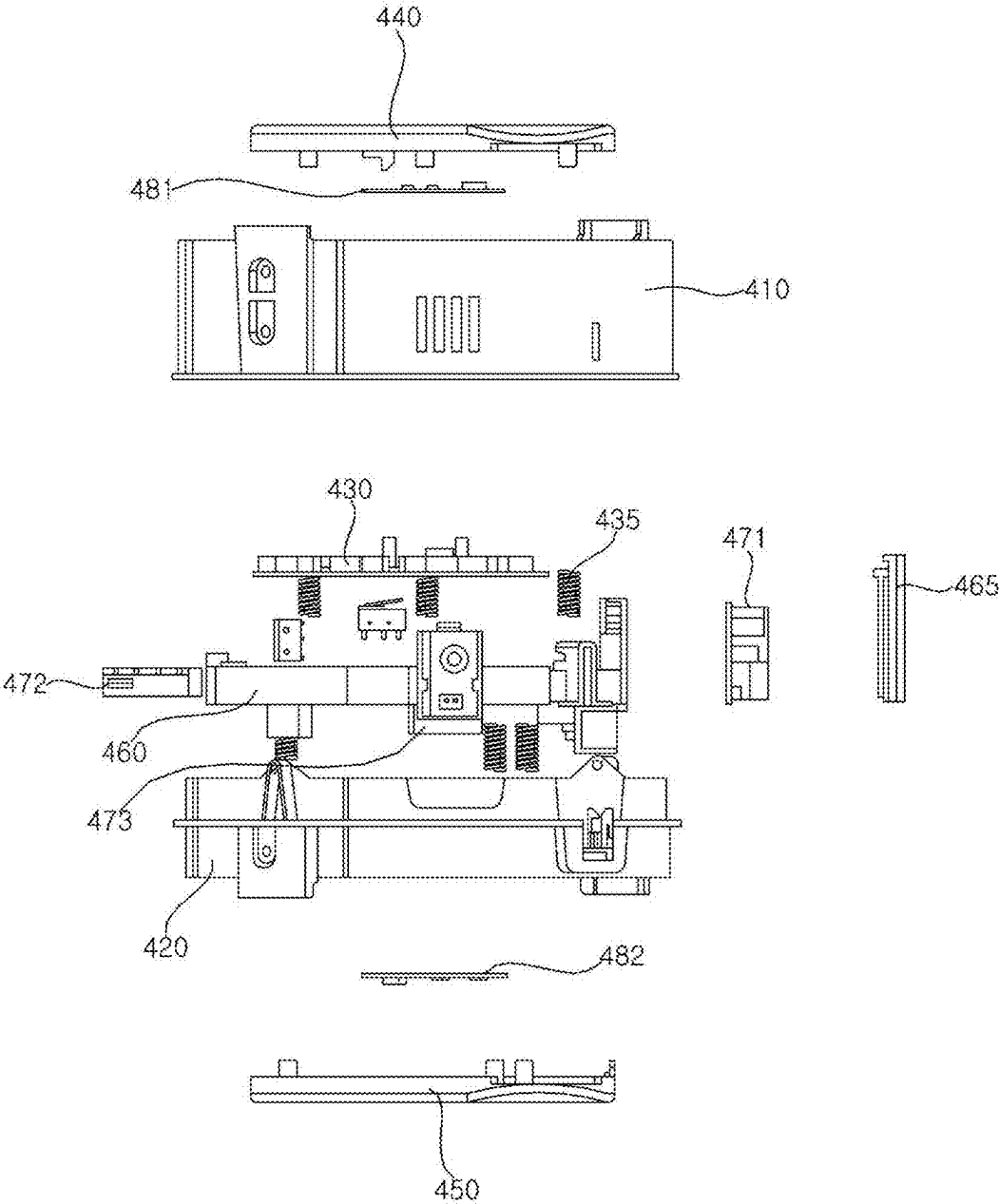


FIG. 18

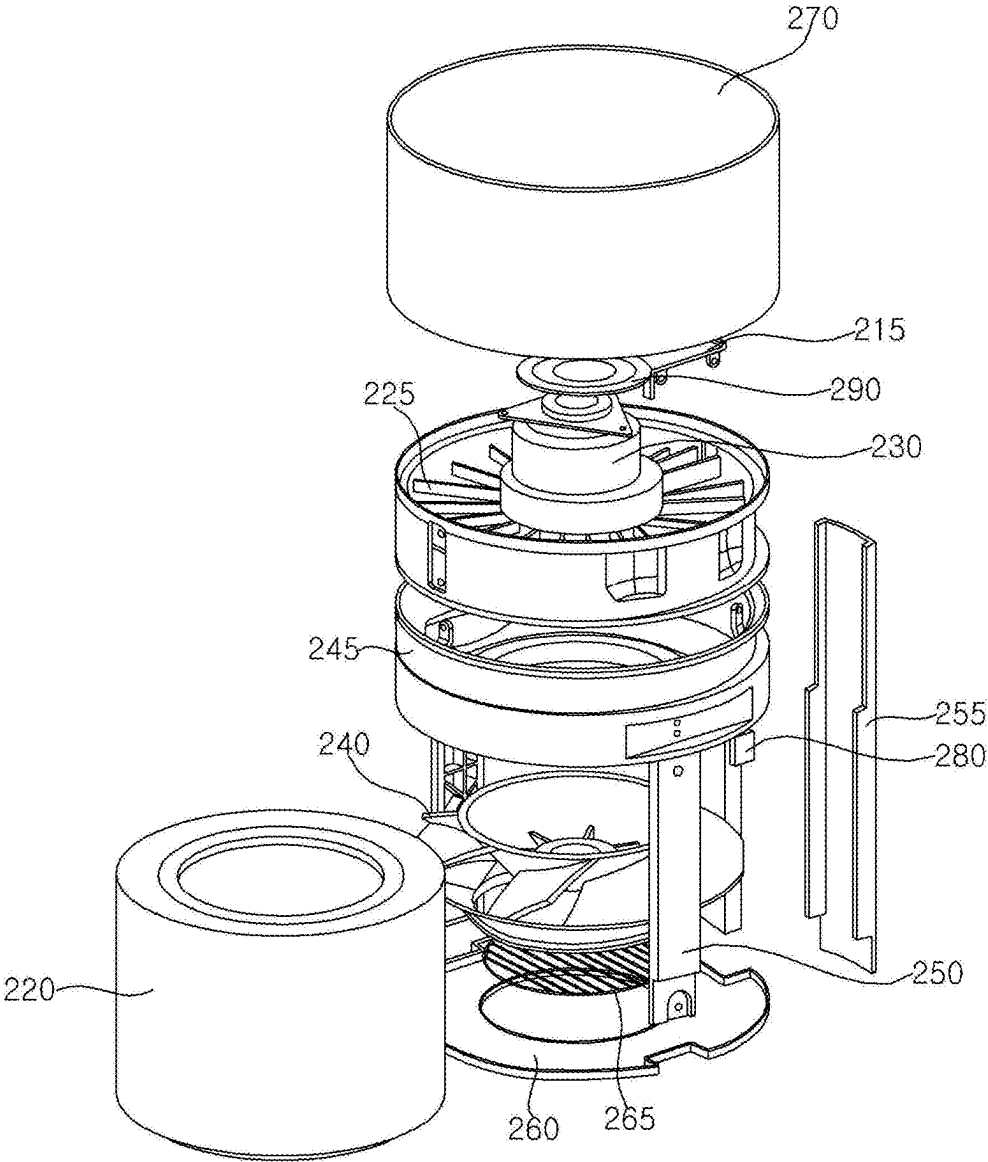
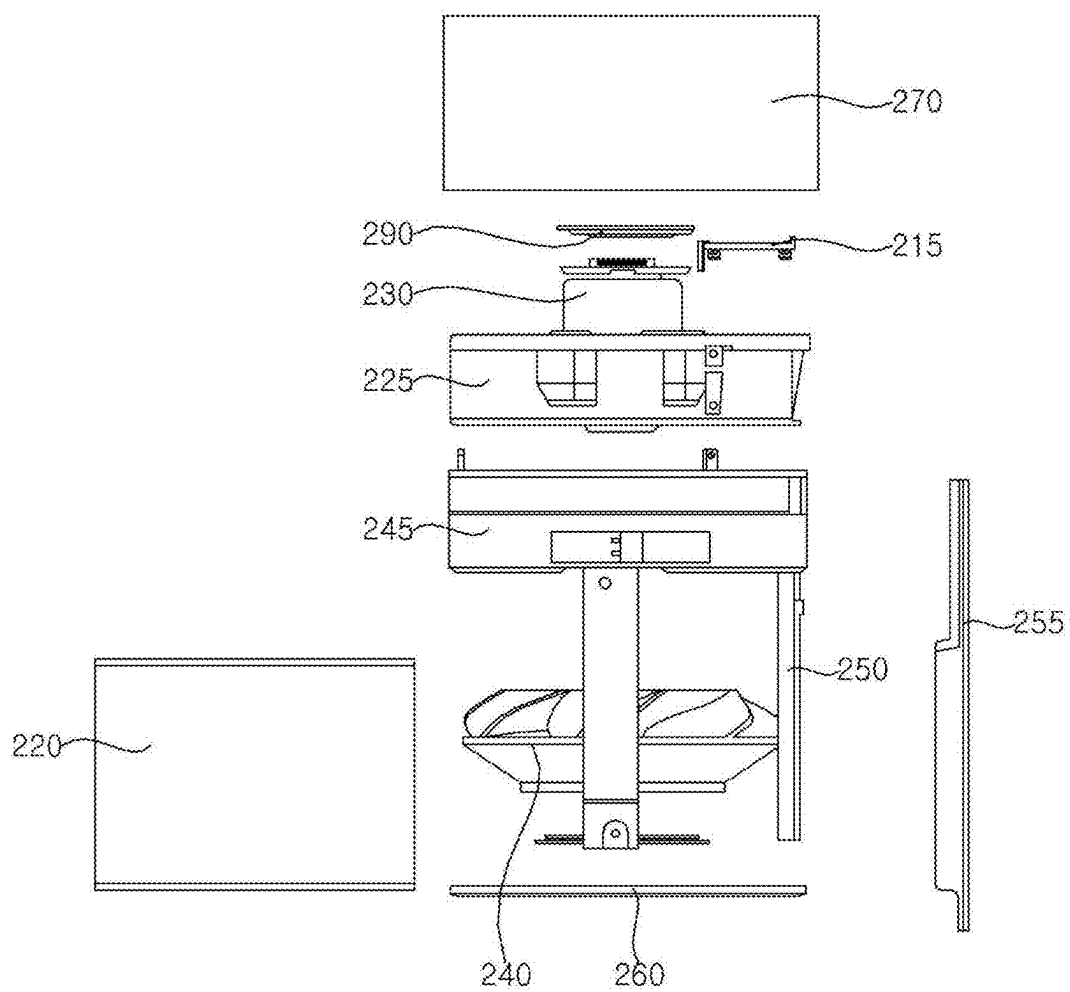


FIG. 19



AIR CLEANER

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Korean Patent Application No. 10-2024-0033178, filed on Mar. 8, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an air cleaner, and more particularly to an air cleaner capable of discharging purified air in various directions.

2. Description of the Related Art

[0003] An air cleaner is a device that filters air in a certain space and discharges the filtered air, thereby reducing the concentration of dust or bacteria in the air in the corresponding space. The air cleaner filters out foreign matter by generating a flow of air in the corresponding space, and discharges air with the foreign matter removed therefrom.

[0004] In order to quickly purify air in the indoor space, a discharge port may be formed in an upper portion of an air cleaner, and filtered air discharged upwardly may be guided so as to flow in all directions.

[0005] Further, in the case in which a discharge port is formed in an upper portion of an air cleaner, a separate fan for regulation of the direction of air discharged may be disposed at an upper portion of the air cleaner in order to cause filtered air to flow a long distance.

[0006] Research on a multistage air cleaner in which blowing devices are stacked vertically is underway.

[0007] A related art document (Korean Patent Laid-Open Publication No. 10-2022-0007365) relates to a circulator for an air cleaner. Such a related art document discloses an air cleaner in which a circulator is disposed above blowing devices disposed in an upward-downward direction in order to cause purified air to flow a long distance.

[0008] However, the air cleaner disclosed in the above related art document has a limitation in that the circulator causes only a portion of purified air discharged from the upper blowing device to flow a long distance.

[0009] Further, the air cleaner disclosed in the above related art document has a problem of requiring an additional fan in order to send purified air away.

SUMMARY OF THE INVENTION

[0010] An aspect of the present disclosure is directed to providing an air cleaner capable of causing discharged purified air to circulate over a longer distance.

[0011] Another aspect of the present disclosure is directed to providing an air cleaner capable of sending a larger amount of purified air away.

[0012] A further aspect of the present disclosure is directed to providing an air cleaner capable of creating various types of airflows.

[0013] Still another aspect of the present disclosure is directed to providing an air cleaner capable of preventing discharged purified air from flowing back thereinto, thereby improving air cleaning performance.

[0014] Still another aspect of the present disclosure is directed to providing an air cleaner capable of preventing discharged purified air from flowing back thereinto without the necessity to add a structure such as a partition plate, thereby having a reduced size.

[0015] Still another aspect of the present disclosure is directed to providing an air cleaner capable of forming an airflow that effectively removes allergens causing allergies.

[0016] Still another aspect of the present disclosure is directed to providing an air cleaner capable of improving suction capacity of air suctioned into the air cleaner.

[0017] The aspects of the present disclosure are not limited to the aspects mentioned above, and other aspects not mentioned herein will be clearly understood by those skilled in the art from the following description.

[0018] An air cleaner according to an embodiment of the present disclosure may deliver all of purified air discharged upwardly to a booster module, thereby sending a larger amount of purified air away.

[0019] An air cleaner according to an embodiment of the present disclosure may create various types of airflows without adding a separate fan by moving a display panel.

[0020] An air cleaner according to an embodiment of the present disclosure may create various types of airflows without adding a separate fan by rotating a booster module.

[0021] In an air cleaner according to an embodiment of the present disclosure, a booster module disposed above an air cleaning module includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, and a display panel disposed in front of the inner grille.

[0022] The booster module includes a motor configured to rotate the head cover.

[0023] The booster module includes a motor configured to move the display panel.

[0024] An inner flow guide is formed inside the head cover. The inner flow guide connects a front edge of the head cover to the inner grille. Air discharged through the inner grille may pass through a slit formed between the inner flow guide and the display panel.

[0025] An air cleaner according to an embodiment of the present disclosure includes a case having a suction port formed in a peripheral surface thereof, an air cleaning module including a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port, and a booster module disposed above the air cleaning module. The booster module includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction, whereby various types of airflows may be created.

[0026] The booster module may further include a first gear connected to and rotated by the first motor and a second gear disposed on an inner surface of the front drive part and formed to be moved in cooperation with the first gear.

[0027] The booster module may further include a first booster motor support part configured to support the first motor and a booster neck configured to accommodate the first booster motor support part and support the inner grille from below. The booster neck may have an upper end formed so as to be inclined.

[0028] The display panel may be disposed parallel to the upper end of the booster neck.

[0029] The head cover may have a hemispherical shape including a first opening in which the display panel is disposed and a second opening in which the booster neck is disposed.

[0030] The first direction may be a direction perpendicular to a plurality of grilles interconnecting an inner wall of the inner grille and an outer wall of the inner grille.

[0031] The booster module may further include a second motor configured to rotate the head cover in a second direction, a third gear connected to and rotated by the second motor, and a fourth gear disposed on an inner surface of the head cover and formed to be moved in cooperation with the third gear. The second direction may be different from the first direction.

[0032] The air cleaner may further include an inner flow guide formed inside the head cover, and the inner flow guide may connect a front edge of the head cover to the inner grille.

[0033] Air discharged through the inner grille may pass through a slit formed between the inner flow guide and the display panel.

[0034] The air cleaning module may include a first air cleaning module and a second air cleaning module disposed above the first air cleaning module.

[0035] Air purified by the second air cleaning module may be discharged through the inner grille.

[0036] The first air cleaning module may include a first discharge port formed in the bottom thereof, and air purified by the first air cleaning module may be discharged through the first discharge port formed in the bottom of the first air cleaning module.

[0037] The air cleaner may further include a bottom plate disposed below the first air cleaning module so as to be in contact with the floor, and an airflow guide may be disposed on the upper surface of the bottom plate so as to guide air discharged downwardly through the first discharge port in a lateral direction.

[0038] The airflow guide may include a curved portion, a flat portion extending laterally from the curved portion, and a boss protruding upward from an end portion of the flat portion.

[0039] A lower discharge grille including a plurality of grilles bent in an outward direction may be disposed inside the first discharge port.

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] The above and other objects, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0041] FIG. 1 is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure;

[0042] FIG. 2A is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure;

[0043] FIG. 2B is an enlarged view of an upper end portion of the air cleaner shown in FIG. 2A;

[0044] FIGS. 3 and 4 are views for explaining operation of a booster module of the air cleaner according to the embodiment of the present disclosure;

[0045] FIGS. 5 and 6 are exploded views of the booster module of the air cleaner according to the embodiment of the present disclosure;

[0046] FIGS. 7 to 10 are cross-sectional views showing the operation modes of the booster module according to the embodiment of the present disclosure;

[0047] FIG. 11 is an assembled view of a first air cleaning module according to the embodiment of the present disclosure;

[0048] FIGS. 12 and 13 are exploded views of the first air cleaning module according to the embodiment of the present disclosure;

[0049] FIG. 14 is an enlarged view of a lower discharge airflow guide structure;

[0050] FIG. 15 is an assembled view of a control module according to the embodiment of the present disclosure;

[0051] FIGS. 16 and 17 are exploded views of the control module according to the embodiment of the present disclosure; and

[0052] FIGS. 18 and 19 are exploded views of a second air cleaning module according to the embodiment of the present disclosure.

DETAILED DESCRIPTION

[0053] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The present disclosure may, however, be embodied in many different forms, and should not be construed as being limited to the embodiments set forth herein.

[0054] In the drawings, illustration of parts unrelated to the description is omitted to clearly and briefly describe the present disclosure, and the same or extremely similar components are denoted by the same reference numerals throughout the specification.

[0055] As used herein, the terms with which the names of components are suffixed, “module” and “unit”, are assigned to facilitate preparation of this specification, and are not intended to suggest unique meanings or functions. Accordingly, the terms “module” and “unit” may be used interchangeably.

[0056] It will be understood that although the terms “first”, “second”, etc., may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another component.

[0057] FIG. 1 is a view showing the external appearance of an air cleaner according to an embodiment of the present disclosure, FIG. 2A is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure, and FIG. 2B is an enlarged view of an upper end portion of the air cleaner shown in FIG. 2A.

[0058] FIGS. 3 and 4 are views for explaining operation of a booster module of the air cleaner according to the embodiment of the present disclosure, and FIGS. 5 and 6 are exploded views of the booster module of the air cleaner according to the embodiment of the present disclosure.

[0059] First, referring to FIGS. 1, 2A, and 2B, a case 10 may have a suction port 10a formed in the peripheral surface thereof to suction air from various directions. Air may be suctioned from all directions with respect to a central line passing through the internal center of the case 10 in an upward-downward direction. The suction port 10a is provided in plural, and the plurality of suction ports 10a is

spaced apart from each other in the circumferential direction of the case 10. The plurality of suction ports 10a is evenly formed in the circumferential direction along the peripheral surface of the case 10 so that air is capable of being suctioned into the case 10 from any direction.

[0060] In this specification, the upward-downward direction or the vertical direction is defined as an axial direction. The axial direction may correspond to the central axis-direction of blower fans 140 and 240, which will be described later, i.e., the motor shaft direction of the fans. The radial direction or the horizontal direction may be understood as a direction perpendicular to the axial direction. The circumferential direction may be understood as a circumferential direction of an imaginary circle that is formed when rotating about the axial direction with a distance in the radial direction as a rotational radius.

[0061] The air cleaner 1 includes air cleaning modules 100 and 200 that generate airflows. The air cleaning modules 100 and 200 include blower fans 140 and 240, which are disposed in the case 10, and filters 120 and 220, which remove foreign matter from air introduced through the suction port 10a, respectively.

[0062] The suction port 10a allows the inside of the case 10 and the outside to communicate with each other. The suction port 10a is provided in plural. The plurality of suction ports 10a may be formed in the peripheral surface of the case 10.

[0063] The plurality of suction ports 10a is evenly formed in the circumferential direction along the outer circumferential surface of the case 10 so that air is capable of being suctioned into the case 10 from any direction.

[0064] The plurality of suction ports 10a is formed in a stripe shape that is elongated in the upward-downward direction. Alternatively, the plurality of suction ports 10a may be formed through perforation in a circular or elliptical shape.

[0065] As described above, because the case 10 is formed in a cylindrical shape and the plurality of suction ports 10a is formed along the outer circumferential surface of the case 10, the amount of air suctioned may increase.

[0066] The air introduced into the suction ports 10a may pass through the filters 120 and 220. Each of the filters 120 and 220 may be formed in a cylindrical shape and may have a filter surface that filters air.

[0067] The case 10 may have a cylindrical shape or a shape of a truncated cone (a cone with the top cut off). A top cover 270 is disposed on the top of the case 10. A booster module 300 is disposed on the top cover 270.

[0068] Referring to FIGS. 1 to 6, the booster module 300 may be coupled to the top cover 270 so as to seal the tops of the air cleaning modules 100 and 200. The top cover 270 may be formed in a shape of a cylinder having a cavity formed in the central portion thereof. A portion of the booster module 300 may be inserted into the central cavity in the top cover 270 from above to below.

[0069] Referring to FIGS. 1 and 2A, the air cleaning modules 100 and 200 include a first air cleaning module 100 and a second air cleaning module 200 that are disposed in the upward-downward direction.

[0070] For example, the second air cleaning module 200 may be disposed on the first air cleaning module 100. Because the first air cleaning module 100 is disposed at a lower portion of the air cleaner 1, the first air cleaning module 100 may be referred to as a “lower air cleaning

module” or a “lower module”, and because the second air cleaning module 200 is disposed at an upper portion of the air cleaner 1, the second air cleaning module 200 may be referred to as an “upper air cleaning module” or an “upper module”.

[0071] The first air cleaning module 100 and the second air cleaning module 200 include blower fans 140 and 240 and filters 120 and 220 configured to remove foreign matter from air introduced through the suction ports 10a, respectively.

[0072] The first air cleaning module 100 includes a first blower fan 140 and a first filter 120 configured to remove foreign matter from air introduced through the suction ports. In addition, the first air cleaning module 100 further includes a first fan motor 130 configured to rotate the first blower fan 140.

[0073] The second air cleaning module 200 includes a second blower fan 240 and a second filter 220 configured to remove foreign matter from air introduced through the suction ports. In addition, the second air cleaning module 200 further includes a second fan motor 230 configured to rotate the second blower fan 240.

[0074] The air cleaning modules 100 and 200 include a discharge port 110a formed in the bottoms thereof. The discharge port 110a is formed below the first blower fan 140 and the first filter 120. In the case in which the second air cleaning module 200 is disposed above the first air cleaning module 100, the discharge port 110a is formed in the bottom of the first air cleaning module 100.

[0075] Air with foreign matter removed therefrom by the first filter 120 is discharged downwardly through the discharge port 110a. Because the air purified in the first air cleaning module 100 is discharged through the discharge port 110a, the discharge port 110a may be referred to as a first discharge port 110a.

[0076] The air cleaner 1 includes a discharge port 310a formed in the booster module 300. The discharge port 310a is formed inside the booster module 300. In addition, the discharge port 310a is formed above the second blower fan 240 and the second filter 220. That is, the discharge port 310a is formed above the second air cleaning module 200.

[0077] Air with foreign matter removed therefrom by the second filter 220 is discharged upwardly through the discharge port 310a. Because the air purified in the second air cleaning module 200 is discharged through the discharge port 310a, the discharge port 310a may be referred to as a second discharge port 310a.

[0078] In addition, a lower discharge grille 110 including a plurality of grilles is disposed at the bottom of the first air cleaning module 100 in order to guide a lower airflow. In addition, an inner grille 310 including a plurality of grilles is disposed inside the booster module 300 in order to guide an upper airflow.

[0079] The inner grille 310 may include an inner wall 311 and an outer wall 312, which have a cylindrical shape. The plurality of grilles of the inner grille 310 may interconnect the inner wall 311 and the outer wall 312 and may be disposed in a radial form.

[0080] The lower discharge grille 110 may include a plurality of grilles having a concentric circle structure. The plurality of grilles may be a plurality of circles having the same center and different radii. Each of the plurality of grilles may include a section bent in the radially outward direction. The lower discharge grille 110 may include a

plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0081] A bottom plate **170** is disposed below the first air cleaning module **100**. The bottom plate **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0082] An airflow guide **170a** (refer to FIG. 12) is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction. The bottom plate **170** may further include a base **170b** (refer to FIG. 12) that supports the airflow guide **170a**. The airflow guide **170a** may be formed on the upper surface of the base **170b**. In addition, the airflow guide **170a** and the base **170b** may be coupled to each other to constitute the bottom plate **170**.

[0083] According to the present disclosure, because the booster module **300** is integrated with the upper discharge passage of the air cleaner **1**, all the air flowing upward may pass through the booster module **300**, and accordingly, a larger amount of purified air may be sent away.

[0084] According to the present disclosure, although two air cleaning modules are stacked vertically, air is not discharged upwardly from the lower air cleaning module. Therefore, while suction performance is improved by forming the suction ports **10a** in the peripheral surface of the case **10**, it is possible to prevent air discharged from the lower air cleaning module from flowing back into the upper air cleaning module.

[0085] The 2-stage air cleaner of the related art has a problem in that purified air discharged from the lower air cleaning module flows upward and is suctioned into the upper air cleaning module, whereby air cleaning performance is reduced. In contrast, according to the present disclosure, purified air discharged from the lower air cleaning module flows downward, and thus introduction of the purified air into the upper air cleaning module is prevented, whereby air cleaning performance is improved.

[0086] Further, according to the present disclosure, the air cleaner **1** does not need to have a structure for preventing introduction of purified air discharged therefrom or configuration for discharge, the overall size of the product may be further reduced, and the freedom of design such as placement of internal parts may be improved.

[0087] For example, a control module **400** may be disposed at the center of the air cleaner **1**, thereby reducing the lengths of power lines and signal lines, simplifying a line connection structure, and reducing the overall height of the product. Further, the control module **400** may be disposed between the air cleaning modules **100** and **200** so that the air cleaning modules **100** and **200** are controlled by the single integrated control module **400**. Accordingly, each of the air cleaning modules **100** and **200** does not need to include an individual control module, and thus the overall height of the product may be reduced.

[0088] General allergen particles have a size of about 3 μm to about 100 μm , which is greater than the size of fine dust, and thus tend to settle on the floor. The lower airflow discharged downwardly from the first air cleaning module **100** causes allergen particles to float in the air and thus to be collected by the filters **120** and **220**. In this way, it is possible

to effectively remove allergens causing allergies using the lower airflow discharged downwardly from the first air cleaning module **100**.

[0089] In addition, the air cleaner **1** may include a UVC LED (not shown), which is disposed in the case **10** and outputs ultraviolet in a wavelength band that has a sterilization effect on bacteria, mold, and microorganisms, and an ionizer (not shown), which is disposed in the case **10** and eliminates germs and mold using electricity supplied thereto.

[0090] The fan motors **130** and **230** may be accommodated in motor fastening parts **125** and **225**, respectively. The air cleaning modules **100** and **200** include filter mounting parts **160** and **260** in which the filters **120** and **220** are mounted, respectively. The control module **400** is disposed between the filters **120** and **220**.

[0091] The annular first discharge port **110a** is formed in a lower portion of the first air cleaning module **100**. Air with foreign matter removed therefrom by the first filter **120** is discharged downwardly through the first discharge port **110a**.

[0092] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0093] The first filter **120** may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter **120**. While the air passes through the first filter **120**, impurities such as fine dust may be removed from the air.

[0094] Because the first filter **120** has a cylindrical shape, air is capable of being introduced into the first filter **120** from any direction. Therefore, the air filtering area may increase.

[0095] The first air cleaning module **100** includes a first fan housing **145** disposed below the first filter **120**, a first blower fan **140** rotatably disposed in the first fan housing **145**, and a first fan motor **130** configured to rotate the first blower fan **140**.

[0096] The first blower fan **140** suction air in the axial direction and discharges the air downwardly in the radial direction. The first fan housing **145** may form a fan path that accommodates the first blower fan **140** and guides the air moved by the first blower fan **140** downwardly.

[0097] The second discharge port **310a** is formed in the booster module **300**. The inner grille **310** is disposed inside the second discharge port **310a** in order to change the direction of air flowing upward. For example, the second discharge port **310a** may be formed in a ring shape. The inner grille **310** may be formed in a radial shape.

[0098] Air with foreign matter removed therefrom by the second filter **220** is discharged upwardly through the second discharge port **310a**. The second air cleaning module **200** includes a second fan housing **245** disposed above the second filter **220**, a second blower fan **240** rotatably disposed in the second fan housing **245**, and a second fan motor **230** configured to rotate the second blower fan **240**.

[0099] The second blower fan **240** suction air in the axial direction and discharges the air upwardly in the radial direction. The second fan housing **245** may form a fan path that accommodates the second blower fan **240** and guides the air moved by the second blower fan **240** upwardly.

[0100] The air moved by the second blower fan **240** flows to the booster module **300**.

[0101] The booster module 300 includes a head cover 320 forming the external appearance thereof and a display panel 390 configured to display various information such as an operation mode and indoor fine dust/harmful gas concentration.

[0102] The head cover 320 may have a hollow hemispherical shape. In this specification, the “hemispherical shape” may be a sphere with a portion cut off without being limited to half of a sphere.

[0103] The head cover 320 may form an inner path that accommodates parts and allows air introduced thereinto from the second air cleaning module 200 to move therealong. The head cover 320 may include a plurality of openings 320a and 320b.

[0104] The head cover 320 may include a first opening 320a in which the display panel 390 is disposed and through which air is discharged and a second opening 320b through which air is introduced thereinto from the second air cleaning module 200.

[0105] The first opening 320a may be formed so as to be inclined at a predetermined angle. The first opening 320a may be disposed at a front portion of the head cover 320. Here, the “front” may be a direction in which the display panel 390 that displays information to the user faces, and the “rear” may be a direction opposite the “front”.

[0106] The display panel 390 may be disposed in the first opening 320a and may form the frontal appearance of the product. In addition, the display panel 390 and the head cover 320 may form a slit 351, which is a space through which air passes.

[0107] The second opening 320b may be formed in a lower portion of the head cover 320. Accordingly, air is introduced into the head cover 320 from the second air cleaning module 200 through the second opening 320b. The air introduced into the head cover 320 through the second opening 320b is discharged through the slit 351, thereby forming the upper airflow.

[0108] The ring-shaped inner grille 310 having a central cavity formed therein and forming the second discharge port 310a is disposed inside the head cover 320. The display panel 390 is disposed in front of the inner grille 310.

[0109] The air introduced into the head cover 320 through the second opening 320b sequentially passes through the inner grille 310 and the slit 351.

[0110] A front drive part 380 may be connected to the display panel 390 through a central cavity 313 in the inner grille 310. The booster module 300 includes a first motor 330 configured to move the display panel 390 in a first direction.

[0111] A booster neck 360 may support the inner grille 310 from below. In addition, the booster neck 360 may support the front drive part 380. The booster neck 360 may be located at the center of the second opening 320b.

[0112] The booster neck 360 may be located above the second fan motor 230 so as not to interfere with the internal airflow. The booster neck 360 is coupled to the air cleaning modules 100 and 200, which are main parts of the air cleaner 1.

[0113] An upper portion of the booster neck 360 may be coupled to the booster module 300. The upper portion of the booster neck 360 may be connected to an inner side of the inner grille 310. The booster neck 360 is coupled to the inner side of the inner grille 310 in order not to block the airflow

discharged from the inner grille 310. That is, the booster neck 360 may be coupled to the inner wall 311 of the inner grille 310.

[0114] A first booster motor support part 335 that supports the first motor 330 is accommodated inside the booster neck 360. The first motor 330 may be coupled to the first booster motor support part 335. Because the first motor 330 is also located inside the booster neck 360, the first motor 330 does not block the inner path extending to the discharge port 310a of the inner grille 310.

[0115] The first motor 330 generates driving force for movement of the display panel 390. For example, the first motor 330 may be a stepper motor, the rotational angle of which is easily controlled. In addition, a motor capable of rotating bidirectionally may be included in the first motor 330.

[0116] The lower end of the booster neck 360 may be coupled to another mechanical structure of the second air cleaning module 200. For example, the lower end of the booster neck 360 may be coupled to an upper side of the second fan housing 245. Alternatively, the lower end of the booster neck 360 may be coupled to an upper side of the second motor fastening part 225. In another embodiment, a separate coupling part (not shown) may be provided between the lower end of the booster neck 360 and the upper end of the second fan housing 245 or the upper end of the second motor fastening part 225.

[0117] The front drive part 380 may be connected to the display panel 390, and accordingly, at least a portion thereof may be moved along with movement of the display panel 390.

[0118] The upper end of the booster neck 360 may be formed so as to be inclined. Accordingly, the inner grille may also be disposed so as to be inclined. In addition, the display panel 390 may be disposed parallel to the upper end of the booster neck 360. Accordingly, it is possible to increase the visibility of information displayed on the display panel 390 and to send purified air, discharged through the slit 351 between the display panel 390 and the head cover 320, away.

[0119] A display base 350 may support the display panel 390 from below, and may have a space defined therein so as to accommodate a printed circuit board. The display base 350 may be moved together with the display panel 390. Therefore, the movement of the display panel 390 may mean movement of the display base 350.

[0120] The slit 351 through which air passes may be formed between the lower surface of the display base 350 and the head cover 320. Accordingly, the display base 350 may guide air discharged through the second discharge port 310a in the radially outward direction.

[0121] The first motor 330 may move the display panel 390 in the first direction, thereby changing at least one of the shape, size, or angle of the slit 351. The form of the discharge airflow may be varied depending on movement of the display panel 390.

[0122] Depending on the movement direction and movement distance of the display panel 390, the size of a space between the display panel 390 and the head cover 320 may be varied, and the flowing direction and flowing distance of the air discharged from the second air cleaning module 200 may be varied.

[0123] The first direction may be a direction perpendicular to the plurality of grilles interconnecting the inner wall 311

and the outer wall 312 of the inner grille 310. For example, the first direction may be a forward direction or a backward direction.

[0124] A first gear 331 is connected to the first motor 330. The first gear 331 may be coupled to the motor shaft of the first motor 330. The first gear 331 may be connected to and rotated by the first motor 330.

[0125] In addition, a second gear 381 extending in the upward-downward direction is disposed on the inner surface of the front drive part 380.

[0126] The first gear 331 is tooth-engaged with the second gear 381. The second gear 381 may operate in cooperation with the first gear 331 and may extend in the forward direction. The second gear 381 is a guide rail that converts rotational motion into linear motion and includes a plurality of gear teeth. The first gear 331 also includes a plurality of gear teeth. The second gear 381 may be a type of rack gear including a plurality of gear teeth disposed in the longitudinal direction thereof. When the first motor 330 is driven, the first gear 331 rotates while being engaged with the second gear 381.

[0127] As the first motor 330 is driven, the first gear 331 is rotated, and the second gear 381 is moved in cooperation therewith. If the first gear 331 is rotated in a predetermined direction by the first motor 330, the second gear 381 of the front drive part 380 may be moved upward. Conversely, if the first gear 331 is rotated in the reverse direction by the first motor 330, the second gear 381 of the front drive part 380 may be moved downward.

[0128] In some embodiments, the front drive part 380 may include a coupling portion (not shown), which is connected to the booster neck 360, and a moving portion (not shown), on which the second gear 381 is mounted and which is connected to the coupling portion and the display base 350. As the first motor 330 is driven, the moving portion may move the display panel 390 and the display base 350 while moving forward or backward.

[0129] FIG. 3 is a view illustrating a state in which the display panel 390 is moved forward, and FIG. 4 is a view illustrating a state in which the display panel 390 is moved backward.

[0130] The first motor 330 may move the display panel 390 forward and backward and may control the form of the airflow discharged from the booster module 300. For example, when the display panel 390 is moved forward, air discharged through the slit 351 may move forward a long distance. When the display panel 390 is moved backward, air discharged through the slit 351 may move in the lateral direction.

[0131] According to the present disclosure, it is possible to create various types of airflows without adding a fan, and because the gear structure does not block the airflow path, air cleaning performance may be improved.

[0132] An inner flow guide 321 is formed inside the head cover 320. The inner flow guide 321 may connect a front edge 322 of the head cover 320 to the inner grille 310.

[0133] The inner surface of the inner flow guide 321 may be formed in a curved shape in order to reduce flow resistance and change the flow direction of air discharged through the discharge port 310a of the inner grille 310.

[0134] The slit 351 is formed between the inner flow guide 350 and the display panel 390. Air discharged through the discharge port 310a of the inner grille 310 passes through the slit 351 and is discharged from the booster module 300.

[0135] The booster module 300 may include a second motor 340 configured to rotate the head cover 320 in a second direction. The second motor 340 may be a stepper motor, the rotational angle of which is easily controlled. In addition, a motor capable of rotating bidirectionally may be included in the second motor 340. The second motor 340 may be coupled to a second booster motor support part 345.

[0136] A third gear 341 is connected to the second motor 340. The third gear 341 may be coupled to the motor shaft of the second motor 340. The third gear 341 may be connected to and rotated by the second motor 340.

[0137] In addition, a fourth gear 325 is disposed on the inner surface of the head cover 320 so as to extend along the inner surface of the head cover 320. The head cover 320 may include a curved portion 323 extending downward from the front edge 322 thereof. The fourth gear 325 may be disposed at the lower end of the inner surface of the curved portion 323.

[0138] The third gear 341 is tooth-engaged with the fourth gear 325. The fourth gear 325 may operate in cooperation with the third gear 341 and may extend in the forward direction. The fourth gear 325 is a guide rail that converts rotational motion into linear motion and includes a plurality of gear teeth. The third gear 341 also includes a plurality of gear teeth. The fourth gear 325 may be a type of rack gear including a plurality of gear teeth disposed in the longitudinal direction thereof. When the second motor 340 is driven, the third gear 341 rotates while being engaged with the fourth gear 325.

[0139] As the second motor 340 is driven, the third gear 341 is rotated, and the fourth gear 325 is moved in cooperation therewith. If the third gear 341 is rotated in a predetermined direction by the second motor 340, the fourth gear 325 of the head cover 320 may be moved upward. Conversely, if the third gear 341 is rotated in the reverse direction by the second motor 340, the fourth gear 325 of the head cover 320 may be moved downward.

[0140] As the second motor 340 is driven, the head cover 320 may be rotated in the second direction. The second direction may be different from the first direction. Accordingly, it is possible to create a wider variety of airflows.

[0141] In general, the air cleaner of the related art is provided with an additional fan in order to send purified air away. Further, because the air cleaner discharge path and the clean booster path are separated from each other, only a portion of the purified air discharged from the air cleaner is sent away via the fan.

[0142] In contrast, according to the present disclosure, because the air cleaner discharge path and the clean booster path are integrated with each other, it is possible to send purified air away without a separate fan and to control the direction and form of the discharge airflow by moving the display panel 390.

[0143] FIGS. 7 to 10 are cross-sectional views showing the operation modes of the booster module according to the embodiment of the present disclosure.

[0144] Referring to FIG. 7, as the first motor 330 is driven, the first gear 331 is rotated, and the second gear 381 is moved forward. The second gear 381 is disposed inside the front drive part 380, and the display panel 390 and the display base 350 are connected to the front drive part 380. Therefore, the display panel 390 is moved forward in accordance with the movement of the second gear 381.

[0145] When the display panel 390 is moved forward, the slit 351 between the display panel 390 and the head cover 320 is widened sufficiently so as to allow air to travel forward. Therefore, when the display panel 390 is moved forward, a front airflow is formed.

[0146] Referring to FIG. 8, as the first motor 330 is driven, the first gear 331 is rotated, and the second gear 381 is moved backward. The display panel 390 is moved backward in accordance with the movement of the second gear 381.

[0147] When the display panel 390 is moved backward, the slit 351 between the display panel 390 and the head cover 320 is narrowed. Therefore, when the display panel 390 is moved backward, a lateral airflow is formed along the narrowed slit 351.

[0148] Referring to FIG. 9, as the second motor 340 is driven, the third gear 341 is rotated in the clockwise direction, and the fourth gear 325 is rotated in the clockwise direction when viewed from the side. The head cover 320 may be driven downward in accordance with the clockwise rotation of the fourth gear 325.

[0149] Referring to FIG. 10, as the second motor 340 is driven, the third gear 341 is rotated in the counterclockwise direction, and the fourth gear 325 is rotated in the counterclockwise direction when viewed from the side. The head cover 320 may be driven upward in accordance with the counterclockwise rotation of the fourth gear 325.

[0150] As shown in FIGS. 9 and 10, when the head cover 320 is driven upward and downward, the slits 351a and 351b are formed asymmetrically, thereby creating a wider variety of airflows.

[0151] FIG. 11 is an assembled view of the first air cleaning module according to the embodiment of the present disclosure, and FIGS. 12 and 13 are exploded views of the first air cleaning module according to the embodiment of the present disclosure.

[0152] Referring to FIGS. 11 to 13, the first discharge port 110a that is open downward is formed in a lower portion of the first air cleaning module 100. The lower discharge grille 110 that changes the direction of air flowing downward is disposed inside the first discharge port 110a.

[0153] The annular first discharge port 110a is formed in a lower portion of the first air cleaning module 100. Air with foreign matter removed therefrom by the first filter 120 is discharged in the downward direction through the first discharge port 110a.

[0154] The lower discharge grille 110 may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille 110 may be disposed inside the first discharge port 110a in order to deliver the lower discharge airflow in the lateral direction.

[0155] The first air cleaning module 100 includes a first filter mounting part 160 to which the first filter 120 is mounted. The first filter 120 may be removably mounted to the first filter mounting part 160. The first filter 120 may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter 120. While the air passes through the first filter 120, impurities such as fine dust may be removed from the air.

[0156] Because the first filter 120 has a cylindrical shape, air is capable of being introduced into the first filter 120 from any direction. Therefore, the air filtering area may increase.

[0157] The first filter mounting part 160 may be formed in a cylindrical shape corresponding to the shape of the first filter 120. In the process of mounting the first filter 120, the

first filter 120 may be slidably fitted into the first filter mounting part 160. Conversely, in the process of demounting the first filter 120, the first filter 120 may be slidably withdrawn from the first filter mounting part 160.

[0158] The first air cleaning module 100 includes a first fan housing 145 disposed below the first filter 120, a first blower fan 140 rotatably disposed in the first fan housing 145, and a first fan motor 130 configured to rotate the first blower fan 140. In addition, the first air cleaning module 100 may further include a wire cover 115 that protects a wire connected to the first fan motor 130.

[0159] The first blower fan 140 suctions air in the axial direction and discharges the air downwardly in the radial direction. The first fan housing 145 may form a fan path that accommodates the first blower fan 140 and guides the air moved by the first blower fan 140 downwardly.

[0160] The first fan motor 130 may be supported by the first motor fastening part 125. The rotation shaft of the first fan motor 130 may extend upward from the first fan motor 130 and may pass through the bottom of the first motor fastening part 125 to be connected to the first blower fan 140.

[0161] First magnets 180 are disposed on two opposite sides of the first fan housing 145. The first magnets 180 may be coupled to metallic pieces of the case 10. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the first fan housing 145, and the magnets may be disposed at the case 10.

[0162] A plurality of support portions 150 extends upward from the first fan housing 145. For example, the support portions 150 may be beams elongated in the longitudinal direction. A wire cover 155 may be disposed on at least one surface of each of the plurality of support portions 150 so as to cover the support portions 150. The support portions 150 and the wire cover 155 may be spaced apart from each other to define a space in which wires or the like are disposed.

[0163] A steel net 165 is disposed between the first filter 120 and the first blower fan 140 in order to prevent body parts of the user or any other object from entering the space in which the first blower fan 140 is disposed.

[0164] The bottom plate 170 is disposed below the first air cleaning module 100. The bottom plate 170 is disposed in contact with the floor to support the air cleaning modules 100 and 200.

[0165] The airflow guide 170a is disposed on the upper surface of the bottom plate 170 in order to guide air discharged downwardly through the first discharge port 110a in the lateral direction. The bottom plate 170 may further include a base 170b that supports the airflow guide 170a.

[0166] FIG. 14 is an enlarged view of the lower discharge airflow guide structure.

[0167] Referring to FIGS. 11 to 14, the first discharge port 110a is formed in the lower end of the first air cleaning module 100, and the lower discharge grille 110 is disposed inside the first discharge port 110a in order to guide the lower discharge airflow.

[0168] The lower discharge grille 110 may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille 110 may be disposed inside the first discharge port 110a in order to deliver the lower discharge airflow in the lateral direction.

[0169] When the first blower fan **140** is rotated by the first motor **130**, the air purified by the first filter **120** is discharged through the first discharge port **110a** as the lower discharge airflow.

[0170] The airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction.

[0171] General allergen particles, which have a size of about 3 μm to about 100 μm , are larger and heavier than indoor floating dust particles, which have a size of about 1 μm , and thus tend to settle on the floor. For example, mold particles have a size of about 10 μm , household dust allergen particles have a size of about 15 μm , and pollen particles have a size of about 50 μm .

[0172] Because virus or cigarette smoke particles have a very small size, the same tends to continuously float without settling on the floor. Indoor floating dust particles remain suspended in the air for a very long time. For example, it takes about 9 to 10 hours for indoor floating dust particles to settle from the ceiling to the floor of the room.

[0173] However, under the same conditions, mold particles, household dust allergen particles, and pollen particles float for a short time of several seconds to several minutes and then settle on the floor. Therefore, it is not easy for the air cleaner of the related art to capture allergen particles. In addition, the only way to remove the settled particles in the related art is to use a vacuum cleaner.

[0174] According to the present disclosure, the lower airflow discharged downwardly from the first air cleaning module **100** may cause allergen particles to float in the air, and the floating allergen particles may be captured by the filters **120** and **220**.

[0175] Meanwhile, according to an embodiment of the present disclosure, the speed of the lower discharge airflow may be controlled in response to the terminal velocity of allergen particles. The speed of the lower discharge airflow may be set to be higher than the terminal velocity of allergen particles.

[0176] The rotational speed of the first blower fan **140** may be controlled based on the terminal velocity of allergen particles. The rotational speed of the first blower fan **140** may correspond to the speed of the lower discharge airflow set to be higher than the terminal velocity of allergen particles. A controller mounted on a main printed circuit board (PCB) **430** of the control module **400** may control the rotational speed of the first blower fan **140**.

[0177] The terminal velocity of allergen particles may be calculated using balance between gravity, buoyancy, and drag force acting on allergen particles. For example, the terminal velocity of allergen particles may be calculated using Stokes' terminal velocity formula.

[0178] In order to cause mold particles having a size of about 10 μm and household dust allergen particles having a size of about 15 μm to float in the air, the speed of the lower discharge airflow may be set to be higher than the terminal velocity of larger household dust allergen particles. For example, the speed of the lower discharge airflow may be set to 0.0038 m/s or higher.

[0179] The airflow guide **170a** includes a curved portion **173**, a flat portion **171** extending laterally from the curved portion **173**, and a boss **172** protruding upward from an end portion of the flat portion **171**.

[0180] The boss **172** may form an airflow that effectively causes allergen particles to float in the air. The boss **172** may reduce flow resistance at the end portion of the flat portion **171**. The boss **172** may prevent flow separation at the end portion of the flat portion **171**. The boss **172** may be formed in a ring shape along the edge of the airflow guide **170a**. The boss **172** may protrude from the flat portion **171** by a predetermined height **d1**, thereby preventing the occurrence of turbulence at the end portion of the flat portion **171** and causing ascending of the airflow. Accordingly, it is possible to effectively cause allergen particles settled on the floor to float in the air while preventing flow separation.

[0181] If the height **d1** of the boss **172** is too great, it may excessively increase the degree of ascending of the airflow, which is inappropriate for causing allergen particles settled on the floor to float in the air. Therefore, it is desirable that the height **d1** of the boss **172** be set to several millimeters (mm) or less, so long as the boss **172** is capable of preventing flow separation.

[0182] The curved portion **173** may be formed so as to surround a central pillar **175** of the bottom plate **170**. The curved portion **173**, which imparts a curvature to the airflow guide **170a**, may be formed so as to be round with a predetermined radius of curvature **R1**. The radius of curvature **R1** may be set so as to guide the lower discharge airflow to smoothly flow in the lateral direction.

[0183] The grille bending angle of the lower discharge grille **110** may be set to 20 degrees to 40 degrees so that the airflow rises at 20 degrees or greater in order to cause allergen particles on the floor to float in the air.

[0184] In addition, the radius of curvature **R1** may be set corresponding to the grille bending angle of the lower discharge grille **110**. For example, a predetermined angle **ag1** based on the vertical direction may be determined corresponding to the grille bending angle of the lower discharge grille **110**, and the radius of curvature **R1** may be set so as to implement the determined angle **ag1**. In more detail, the angle **ag1** may be set to be smaller than the grille bending angle of the lower discharge grille **110**.

[0185] The predetermined angle **ag1** may be an angle between the central pillar **175** and the curved portion **173**. In this case, the predetermined angle **ag1** may be smaller than the angle of the discharge airflow through the discharge port **110**. The angle of the discharge airflow may correspond to the grille arrangement angle and grille bending angle of the lower discharge grille **110**.

[0186] The curved portion **173** may guide the lower discharge airflow to smoothly flow in the lateral direction, thereby reducing flow resistance. According to the present disclosure, in order to deliver the lower discharge airflow in the lateral direction, the bottom plate **170** is imparted with a curvature, and the boss **172** is formed at an end portion of the flat portion **171**. Accordingly, a turbulent flow for causing allergen particles to float in the air may be created. That is, because the airflow guide **170a** is imparted with an inclination and a curvature and is provided with the boss for preventing flow separation, resistance to the lower discharge airflow may be minimized.

[0187] FIG. 15 is an assembled view of the control module according to the embodiment of the present disclosure, and FIGS. 16 and 17 are exploded views of the control module according to the embodiment of the present disclosure.

[0188] Referring to FIGS. 15 to 17, the control module **400** includes an upper cover **410** and a lower cover **420** that

are coupled to each other to form the external appearance of the control module 400. The control module 400 includes a main PCB 430 on which circuits, such as a controller for control of overall operation of the air cleaner 1, are mounted, and the main PCB 430 is accommodated in an inner space defined by the upper cover 410 and the lower cover 420.

[0189] Each of the upper cover 410 and the lower cover 420 has a central cavity formed therein, and an upper filter pressing plate 440 and a lower filter pressing plate 450 are disposed in the central cavity of the upper cover 410 and the central cavity of the lower cover 420, respectively. Each of the upper filter pressing plate 440 and the lower filter pressing plate 450 may be connected to a plurality of springs 435.

[0190] The control module 400 may be located between the first air cleaning module 100 and the second air cleaning module 200. The control module 400 may be located above the first air cleaning module 100 and below the second air cleaning module 200.

[0191] Because the control module 400 is disposed between the first air cleaning module 100 and the second air cleaning module 200, the lengths of wires connecting the control module 400 to the first and second air cleaning modules 100 and 200 and variation in the lengths of the wires may be minimized, and the wiring structure may be simplified.

[0192] Further, because the filter pressing plates 440 and 450 are integrated with the upper and lower covers 410 and 420, vibration and noise may be further reduced, and manufacturing costs may be reduced.

[0193] When the first filter 120 is mounted to the first filter mounting part 160 of the first air cleaning module 100, the first filter 120 presses the lower filter pressing plate 450. When the second filter 220 is mounted to the second filter mounting part 260 of the second air cleaning module 200, the second filter 220 presses the upper filter pressing plate 440. Accordingly, the filters 120 and 220 may be more securely supported.

[0194] In addition, the control module 400 includes an upper UVC LED 481 and a lower UVC LED 482. The upper UVC LED 481 outputs UVC light upwardly toward the second air cleaning module 200 and the second filter 220. The lower UVC LED 482 outputs UVC light downwardly toward the first air cleaning module 100 and the first filter 120.

[0195] In general, ultraviolet light may be subdivided according to wavelength. For example, ultraviolet light may be classified into UVA having a wavelength of 320 nm to 400 nm, UVB having a wavelength of 280 nm to 320 nm, and UVC having a wavelength of 200 nm to 280 nm. In particular, UVC, which is short-wavelength ultraviolet light, has characteristics of destroying DNA of bacteria and causing a chemical reaction with a specific substance, and thus is effective for sterilization.

[0196] The control module 400 includes a mounting part 460 that supports the main PCB 430. Various sensors, such as an LED sensor 471, a laser sensor 472, and a gas sensor 473, may be mounted to the mounting part 460. Some of the sensors (e.g., the LED sensor 471) may include a cover 465.

[0197] The LED sensor 471 may emit infrared LED light and may detect light scattered by dust, thereby sensing the amount of dust. The laser sensor 472 may sense the amount of dust using laser light. The gas sensor 473 may sense the amount of gas in the air.

[0198] Although the LED sensor 471, the laser sensor 472, and the gas sensor 473 are illustrated in FIGS. 16 and 17, the disclosure is not limited thereto. The types of sensors provided and combinations thereof may be varied. In addition, one or more communication modules (not shown), such as a Wi-Fi module, may be provided inside the control module 400.

[0199] FIGS. 18 and 19 are exploded views of the second air cleaning module according to the embodiment of the present disclosure.

[0200] Referring to FIGS. 18 and 19, a hollow upper cover 270 is disposed at an upper portion of the second air cleaning module 200. Air with foreign matter removed therefrom by the second filter 220 flows upward and enters the booster module 300 through the upper cover 270.

[0201] In the second air cleaning module 200, the second filter 220 is mounted to the second filter mounting part 260. The above description of the first filter 120 and the first filter mounting part 160 may be similarly applied to the second filter 220 and the second filter mounting part 260. However, while the first filter 120 and the first filter mounting part 160 are located above the first blower fan 140 and the first fan housing 145, the second filter 220 and the second filter mounting part 260 are located below the second blower fan 240 and the second fan housing 245.

[0202] The second air cleaning module 200 includes a second fan housing 245 disposed above the second filter 220, a second blower fan 240 rotatably disposed in the second fan housing 245, and a second fan motor 230 configured to rotate the second blower fan 240.

[0203] In addition, the second air cleaning module 200 may further include a wire cover 215 that protects a wire connected to the second fan motor 230. The second air cleaning module 200 may further include a rotary bearing 290 that reduces frictional force generated during rotation.

[0204] The second blower fan 240 suctions air in the axial direction and discharges the air upwardly in the radial direction. The second fan housing 245 may form a fan path that accommodates the second blower fan 240 and guides the air moved by the second blower fan 240 upwardly.

[0205] The second fan motor 230 may be supported by the second motor fastening part 225. The rotation shaft of the second fan motor 230 may extend downward from the second fan motor 230 and may pass through the top of the second motor fastening part 225 to be connected to the second blower fan 240.

[0206] Second magnets 280 are disposed on two opposite sides of the second fan housing 245. The second magnets 280 may be coupled to metallic pieces of the case 10. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the second fan housing 245, and the magnets may be disposed at the case 10.

[0207] A plurality of support portions 250 extends downward from the second fan housing 245. For example, the support portions 250 may be beams elongated in the longitudinal direction. A wire cover 255 may be disposed on at least one surface of each of the plurality of support portions 250 so as to cover the support portions 250. The support portions 250 and the wire cover 255 may be spaced apart from each other to define a space in which wires or the like are disposed.

[0208] A steel net **265** is disposed between the second filter **220** and the second blower fan **240** in order to prevent body parts of the user or any other object from entering the space in which the second blower fan **240** is disposed.

[0209] As is apparent from the above description, according to at least one of the embodiments of the present disclosure, a booster module, which includes a head cover, a ring-shaped inner grille having a central cavity formed therein and forming a discharge port inside the head cover, a display panel disposed in front of the inner grille, a front drive part connected to the display panel through the central cavity of the inner grille, and a first motor configured to move the display panel in a first direction is disposed above an air cleaning module, thereby creating various types of airflows without adding a fan.

[0210] According to at least one of the embodiments of the present disclosure, the booster module further includes a first gear connected to and rotated by the first motor and a second gear disposed on an inner surface of the front drive part and formed to be moved in cooperation with the first gear, whereby the gear structure does not block an airflow path, and thus air cleaning performance may be improved.

[0211] According to at least one of the embodiments of the present disclosure, the upper end of a booster neck supporting the inner grille is formed so as to be inclined, and the display panel is disposed parallel to the upper end of the booster neck, whereby the visibility of information displayed on the display panel may be improved, and purified air may be sent away.

[0212] According to at least one of the embodiments of the present disclosure, the booster module further includes a second motor configured to rotate the head cover in a second direction, a third gear connected to and rotated by the second motor, and a fourth gear disposed on an inner surface of the head cover and formed to be moved in cooperation with the third gear, whereby a wider variety of airflows may be created.

[0213] According to at least one of the embodiments of the present disclosure, an inner flow guide is formed inside the head cover so as to connect a front edge of the head cover to the inner grille, thereby reducing internal flow resistance.

[0214] According to at least one of the embodiments of the present disclosure, the air cleaning module includes a first air cleaning module and a second air cleaning module disposed above the first air cleaning module, thereby improving air cleaning performance.

[0215] According to at least one of the embodiments of the present disclosure, air purified by the second air cleaning module is discharged through the inner grille, whereby all of the air purified by the second air cleaning module may be delivered to the booster module, and thus a larger amount of purified air may be sent away.

[0216] According to at least one of the embodiments of the present disclosure, a first discharge port is formed in the bottom of the first air cleaning module, and air purified by the first air cleaning module is discharged through the first discharge port formed in the bottom of the first air cleaning module, whereby a lower discharge airflow that is effective for removal of allergen particles may be created.

[0217] According to at least one of the embodiments of the present disclosure, an airflow guide is disposed on the upper surface of a bottom plate, which is disposed below the air cleaning module, in order to guide air discharged downwardly through the first discharge port in the lateral direc-

tion, whereby allergens causing allergies may be caused to float in the air and may be captured. Accordingly, allergens may be effectively removed.

[0218] According to at least one of the embodiments of the present disclosure, the airflow guide includes a boss protruding upward from an end portion thereof, thereby reducing flow resistance.

[0219] According to at least one of the embodiments of the present disclosure, a lower discharge grille including a plurality of grilles bent in the outward direction is disposed inside the first discharge port, thereby creating a lower discharge airflow that causes allergen particles to float in the air.

[0220] Although the present disclosure has been described with reference to specific embodiments shown in the drawings, it is apparent to those skilled in the art that the present disclosure is not limited to those embodiments and is embodied in many forms without departing from the scope of the present disclosure, which is described in the following claims. These modifications should not be individually understood from the technical spirit or scope of the present disclosure.

What is claimed is:

1. An air cleaner comprising:

a case comprising a suction port disposed on a peripheral surface of the case;

an air cleaning module comprising a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port; and

a booster module disposed above the air cleaning module, the booster module comprising:

a head cover;

a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity;

a display panel disposed in front of the inner grille;

a front drive part connected to the display panel through the central cavity of the inner grille; and

a first motor configured to move the display panel in a first direction.

2. The air cleaner according to claim 1, wherein the booster module further comprises:

a first gear configured to be rotated by the first motor; and

a second gear disposed on an inner surface of the front drive part, the second gear being configured to be moved by the first gear.

3. The air cleaner according to claim 1, wherein the booster module further comprises:

a booster motor support part configured to support the first motor; and

a booster neck configured to accommodate the booster motor support part and support the inner grille.

4. The air cleaner according to claim 3, wherein an upper end of the booster neck is inclined with respect to a vertical direction.

5. The air cleaner according to claim 4, wherein the display panel is disposed to be parallel to the upper end of the booster neck.

6. The air cleaner according to claim 3, wherein the head cover has a curved shape and comprises a first opening configured to include the display panel, and a second opening configured to include the booster neck.

7. The air cleaner according to claim 1, wherein the inner grille comprises a plurality of grilles connecting an inner wall of the inner grille and an outer wall of the inner grille, and

wherein the first direction is a direction perpendicular to a length of the plurality of grilles.

8. The air cleaner according to claim 1, wherein the booster module further comprises:

a second motor configured to rotate the head cover in a second direction;

a third gear configured to be rotated by the second motor; and

a fourth gear disposed on an inner surface of the head cover, the fourth gear being configured to be moved by the third gear.

9. The air cleaner according to claim 8, wherein the second direction is different from the first direction.

10. The air cleaner according to claim 1, further comprising an inner flow guide disposed inside the head cover, the inner flow guide connecting a front edge of the head cover to the inner grille.

11. The air cleaner according to claim 10, wherein the inner grille is configured to discharge air through at least one slit disposed between the inner flow guide and the display panel.

12. The air cleaner according to claim 1, wherein the air cleaning module comprises:

a first air cleaning module; and

a second air cleaning module disposed above the first air cleaning module, and

wherein the inner grille is configured to discharge air filtered by the second air cleaning module.

13. The air cleaner according to claim 12, wherein the first air cleaning module comprises a first discharge port disposed in a bottom of the first air cleaning module, the first discharge port being configured to discharge air filtered by the first air cleaning module.

14. The air cleaner according to claim 13, further comprising a bottom plate disposed below the first air cleaning module, the bottom plate being configured to be in contact with a floor, the bottom plate comprising an airflow guide disposed on an upper surface of the bottom plate, the airflow

guide being configured to guide air discharged downwardly through the first discharge port in a lateral direction.

15. The air cleaner according to claim 14, wherein the airflow guide comprises:

a curved portion;

a flat portion extending laterally from the curved portion; and

a boss protruding upward from an end portion of the flat portion.

16. The air cleaner according to claim 13, further comprising a lower discharge grille disposed inside the first discharge port, the lower discharge grille comprising a plurality of grilles bent in an outward direction.

17. An air cleaner comprising:

a case comprising a suction port disposed on a peripheral surface of the case;

an air cleaning module comprising a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port; and a booster module disposed above the air cleaning module, the booster module comprising:

a head cover;

a discharge port disposed inside the head cover, the discharge port including a ring-shaped inner grille having a central cavity;

a display panel disposed in front of the inner grille; and a motor configured to rotate the head cover.

18. The air cleaner according to claim 17, wherein the booster module further comprises:

a front drive part connected to the display panel through the central cavity of the inner grille; and

a motor configured to move the display panel through the front drive part.

19. The air cleaner according to claim 18, further comprising an inner flow guide disposed inside the head cover, the inner flow guide connecting a front edge of the head cover to the inner grille.

20. The air cleaner according to claim 19, wherein the inner grille is configured to discharge air through at least one slit disposed between the inner flow guide and the display panel.

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